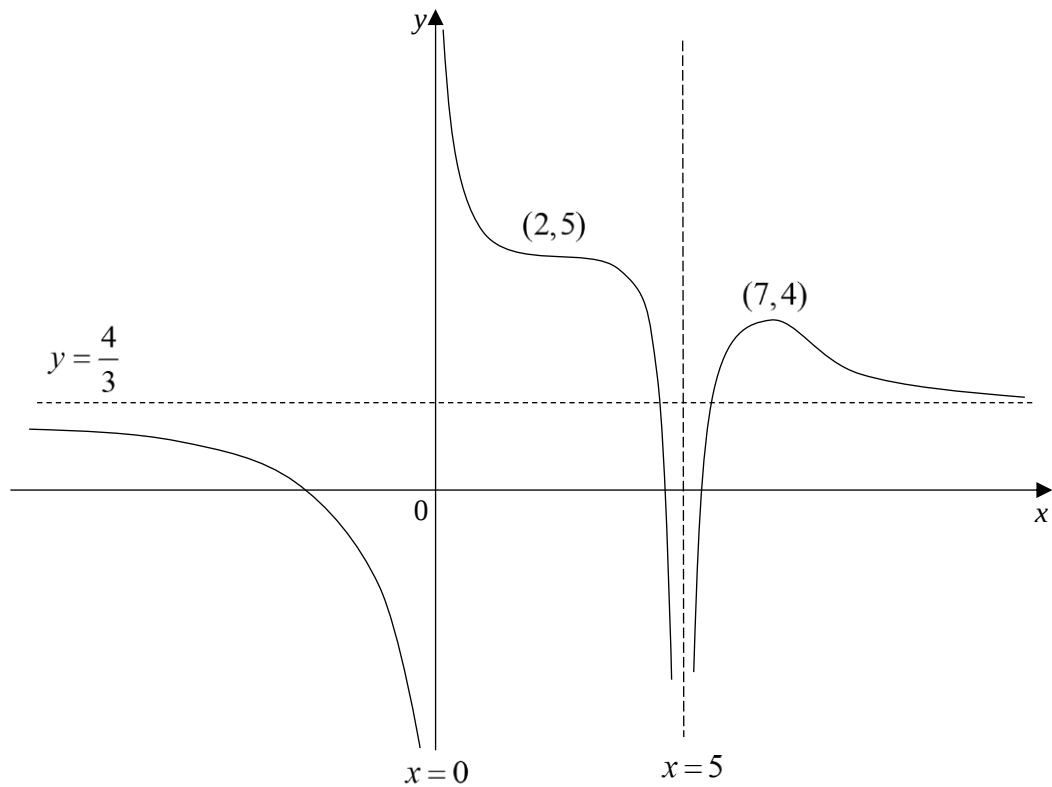


Promo Practice Paper 4 [HCI 2022] 100m

- 1 The diagram below shows the curve $y = f(x)$. The curve has a stationary point of inflexion at $(2,5)$ and a maximum point at $(7,4)$. The curve also has vertical asymptotes $x = 0$, $x = 5$ and a horizontal asymptote $y = \frac{4}{3}$.



Sketch the curve $y = f'(x)$, labelling clearly the coordinates of any axial intercepts, turning points and equations of asymptotes where applicable. [3]

- 2 A local family consisting of adults, children and senior citizens, is planning for a trip to the Jewel Bird Park. The ticket prices are listed as shown below. If all members in the family purchase the Local Resident Discounted tickets, they will need to pay a total price of \$301.20. If all members in the family purchase the Wildlife Quest Bundle, they will need to pay a total price of \$478.50. Given that the number of adults is four times the number of senior citizens, find the number of adults, children and senior citizens in the family. [3]

Local Resident Discounted Tickets	
Adult	\$27.00
Senior Citizen Ages 60 and above	\$15.00
Child Ages 3 to 12	\$18.40

- 3** A curve C has equation $kxe^y + ke^x = y^2 + k^2$, where k is a positive constant.
- (i) Express $\frac{dy}{dx}$ in terms of x, y and k . [3]
- (ii) Explain why there is no point on C where the tangent is parallel to the x -axis. [2]
- 4** With respect to origin O , the distinct points P, Q, R and S have position vectors $\mathbf{p}, \mathbf{q}, \mathbf{r}$ and $\mathbf{s}$ respectively. It is known that $\mathbf{q}$ is a unit vector.
- (i) Given that $(\mathbf{r} - \mathbf{p}) \times (\mathbf{p} - \mathbf{q}) = \mathbf{0}$, state, with justification, the relationship between the points P, Q and R . [2]
- (ii) Give a geometrical interpretation of $|\mathbf{q} \cdot (\mathbf{r} - \mathbf{s})|$. [1]
- (iii) Given that $\mathbf{p} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$ and that $\mathbf{q}$ is parallel to and in the opposite direction of $\mathbf{p}$, find $\mathbf{q}$. [2]

5 **MI Promo 9758/2020/PU2/P1/Q8a**

Do not use a calculator in answering this question.

Given that $a = 3 - i$ and $b = 5 + 2i$, find the following complex numbers in the form $x + iy$,

(i) ab^* , [2]

(ii) $\frac{b}{a^*}$. [3]

6 **RVHS Promo 9758/2020/Q3**

It is given that $\tan^{-1}(y) = 1 - e^{-x}$.

(i) Show that $e^x \frac{dy}{dx} = 1 + y^2$. [1]

(ii) By further differentiation of the expression in part (i), find the Maclaurin series for y , up to and including the term in x^2 . [4]

(iii) Hence, or otherwise, find the Maclaurin series for $\frac{1+y^2}{e^x}$, up to and including the term in x . [1]

7 **ASRJC Promo 9758/2020/Q6**

The curve C_1 whose equation is $x^2 + y^2 = 16$ undergoes, in succession, the following transformations:

- A: Translation of 4 units in the negative x -direction.
- B: Translation of 2 units in the positive y -direction.
- C: Scaling by factor 2 parallel to the y -axis.

(i) Find the equation of the resulting curve C_2 . [2]

(ii) Sketch C_1 and C_2 on the same diagram, indicating clearly the relevant features of the two curves. You need not find the coordinates of axial intercepts. [2]

(iii) State the coordinates of the points of intersection between C_1 and C_2 . [1]

(iv) Find the overlapping area between the two curves C_1 and C_2 . [2]

8 The sum of the first n terms of a series is given by the expression $e^2 - (-2)^n (e^{2-n})$.

(i) State the first term of the series in terms of e . [1]

(ii) By finding the n^{th} term of the series, show that this is a geometric series. [3]

(iii) Explain why the sum to infinity, S , of the series exists, and determine the exact value of S . [3]

- 9 A curve C has parametric equations

$$x = \theta^2 + 1, \quad y = 2 \sin \theta + 1 \quad \text{where } -\pi \leq \theta \leq \pi.$$

- (i) Find the exact values of θ at which C crosses the x -axis. [1]
- (ii) Sketch C , labelling the coordinates of the points at which C crosses the x -axis. [2]
- (iii) Find the equation of the tangent to C at the point P with parameter p , where $-\pi \leq p \leq \pi$. [3]
- (iv) Q is a point on C such that the tangent at Q is parallel to the y -axis. Find the area bounded by C , the tangent at Q and the x -axis. [3]

- 10 A function f is said to be self-inverse if $f(x) = f^{-1}(x)$ for all x in the domain of f . It is given that g is a self-inverse function and is defined by

$$g: x \mapsto \frac{x+a}{3x+b}, \text{ for } x \in \mathbb{R}, \quad x > -\frac{b}{3},$$

where a and b are constants and $g(1) = 5$.

- (i) Find the value of b and show that $a = 9$. [3]
- (ii) Find $g^{2021}(1)$. [1]

The function h is defined by

$$h: x \mapsto |1-x|(x+5), \text{ for } x \in \mathbb{R}, \quad x \geq 2.$$

- (iii) Given that h^{-1} exists, find h^{-1} in similar form. [4]
- (iv) Show that gh exists and find the exact range of gh . [3]

- 11 The curve C has equation $y = \frac{(1-x)(x-2)}{x^2 - 2x - 8}$.

- (i) Find the value of x when $y = -1$. [1]
- (ii) Sketch C , showing clearly the equations of any asymptotes and coordinates of any turning points and axial intercepts. [3]
- (iii) Solve the inequality [1]

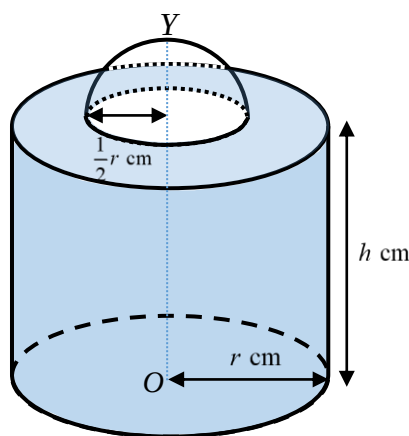
$$\frac{-x^2 + 3x - 2}{x^2 - 2x - 8} \geq -1.$$

- (iv) Using the result in part (iii), solve the inequality [2]

$$\frac{-1 + 3x - 2x^2}{1 - 2x - 8x^2} \geq -1.$$

It is given that a sphere of radius R has surface area $4\pi R^2$ and volume $\frac{4}{3}\pi R^3$.

- 12 A perfume maker designs a prototype of a perfume bottle of fixed volume $V \text{ cm}^3$ for a new fragrance as shown in the diagram below.



The prototype comprises 2 different segments where the vertical axis of the prototype, OY , is $\left(h + \frac{1}{2}r\right) \text{ cm}$. The bottom segment is a glass cylinder of radius $r \text{ cm}$ and height $h \text{ cm}$. The top segment is a chrome-plated plastic hemisphere of radius $\frac{1}{2}r \text{ cm}$. It is assumed that the prototype is of negligible thickness and there is no gap between the 2 segments.

- (i) Find h in terms of V , r and π . [2]
- (ii) Given that the cost of manufacturing the glass cylinder is \$8 per cm^2 and the chrome-plated plastic is \$3 per cm^2 , show that the total cost of manufacturing the prototype is $\$ \left(\frac{85}{6} \pi r^2 + \frac{16V}{r} \right)$. Hence, using differentiation, find the exact value of $\frac{h}{r}$ such that the cost of manufacturing the prototype is minimum. [7]

It is now given that the diameter of the chrome-plated hemisphere is 3 cm and the height of the glass cylinder is 5 cm.

- (iii) A crack at the bottom of the prototype causes the perfume to leak out of the prototype at a constant rate of $1.5 \text{ cm}^3 / \text{s}$. Given that perfume is initially filled to the brim of the top segment of the prototype, find the exact rate of decrease of the height of the perfume in the prototype 5 seconds after the prototype has cracked. [3]

- 13** The points A , B and C have coordinates $(-1, -12, 4)$, $(5, 0, 7)$ and $(6, 1, 4)$ respectively. The line l_1 has equations $\frac{x-1}{2} = \frac{2-y}{3}$, $z = 4$ and the line l_2 passes through A and B .

- (i) Find the coordinates of the foot of perpendicular from C to l_1 . [4]
- (ii) Find the acute angle between l_1 and l_2 . [3]
- (iii) The point D is on l_2 such that the distance from D to A is twice the distance from D to B . Find the possible point(s) D . [4]
- (iv) The line l_3 passes through point A and is perpendicular to both l_1 and l_2 . Find the equation of l_3 . [2]

- 14** A couple takes up a housing loan of $\$L$ and the interest is charged before each monthly repayments at a fixed rate of $p\%$ per annum. Their monthly repayment commences on 1 September 2021. Monthly repayments of $\$x$ are due and payable on the first day of subsequent months until their housing loan is fully repaid.

- (i) State an expression in terms of L and p for the interest charged before their first repayment on 1 September 2021. [1]
- (ii) Show that the outstanding loan at the start of the n^{th} month after their monthly repayment is given by

$$\left(1 + \frac{p}{1200}\right)^n L - \frac{1200x}{p} \left[\left(1 + \frac{p}{1200}\right)^n - 1 \right].$$

[3]

The couple is taking a housing loan of $\$504,000$ at a fixed interest rate of 2.6% per annum.

- (iii) Calculate the monthly repayment if the couple plans to repay the loan in 30 years. [2]
- (iv) Given that the couple decides to pay monthly repayments of $\$4000$, find the date at which the couple will be able to fully repay their housing loan and the amount that the couple pays for their final monthly repayment. [3]

Answers

<u>2</u>	8 adults, 3 children and 2 senior citizens
<u>3</u>	(i) $\frac{dy}{dx} = \frac{ke^x + ke^y}{2y - kxe^y}$
<u>4</u>	(iii) $-\frac{1}{\sqrt{21}} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$
<u>5</u>	(ai) $13 - 11i$, (aii) $\frac{1}{10}(17 + i)$,
<u>6</u>	(ii) $y = x - \frac{1}{2}x^2 + \dots$ (iii) $\frac{1+y^2}{e^x} = 1 - x + \dots$
<u>7</u>	(i) $\frac{(x+4)^2}{4^2} + \frac{(y-4)^2}{8^2} = 1$ (iii) $(-2.32, -3.26)$ and $(0, 4)$ (iv) 19.2
<u>8</u>	(i) $e^2 + 2e$ (ii) $(-2)^{n-1}(e^{2-n})[e+2]$ (iii) e^2
<u>9</u>	(i) $\theta = -\frac{\pi}{6}, -\frac{5\pi}{6}$ (iii) $y = \frac{x \cos p}{p} - \frac{(p^2+1)\cos p}{p} + 2 \sin p + 1$ (iv) 0.0880 units^2
<u>10</u>	(i) $b = -1$. (ii) 5 (iii) $h^{-1}: x \mapsto \sqrt{9+x} - 2, \quad x \in \mathbb{R}, x \geq 7$. (iv) $R_{\text{gh}} = \left(\frac{1}{3}, \frac{4}{5}\right]$
<u>11</u>	(i) 10 (iii) $-2 < x < 4$ or $x \geq 10$ (iv) $-2 < \frac{1}{x} < 4$ or $\frac{1}{x} \geq 10$ (v) $1 + \ln \frac{3}{8} \text{ unit}^2$
<u>12</u>	(i) $h = \frac{V}{\pi r^2} - \frac{r}{12}$ (ii) $\frac{h}{r} = \frac{27}{16}$

	(iii) $\frac{1}{6\pi} \text{ cm / s}$
<u>13</u>	(i) $F(3, -1, 4)$ (ii) 61.0° (iii) $D(3, -4, 6)$ or $D(11, 12, 10)$ (iv) $l_3 : \vec{r} = \begin{pmatrix} -1 \\ -12 \\ 4 \end{pmatrix} + \alpha \begin{pmatrix} -3 \\ -2 \\ 14 \end{pmatrix} \quad \alpha \in \mathbb{R}$
<u>14</u>	(i) $\frac{pL}{1200}$ (iii) \$2017.72. (nearest cent) or \$2020. (3 s.f.) (iv) \$1245.38 (v) $k = 500$, $a = 50$ (vi) 87